

# Special39 / competency practice — Paper 01 — Electricity: Measurement, Evidence and Laboratory Decisions

CLASS 10 • ELECTRICITY • ORIGINAL PRACTICE

## Original Class 10 Electricity • 39 questions • 80 marks • competency practice

Original chapter practice — not an official examination paper, a prediction, or an official chapter weightage.

### 39 questions • 80 marks • 180 minutes

Original Class 10 Electricity chapter practice: 39 questions, 80 marks. Constructed experiments and data support broad chapter coverage. Not an official paper, prediction or claim about chapter weightage.

### Paper structure

Section A | Questions 1–20 |  $20 \times 1 = 20$  marks

Section B | Questions 21–26 |  $6 \times 2 = 12$  marks

Section C | Questions 27–33 |  $7 \times 3 = 21$  marks

Section D | Questions 34–36 |  $3 \times 5 = 15$  marks

Section E | Questions 37–39 |  $3 \times 4 = 12$  marks

### Instructions

Attempt all questions. There are no internal OR choices. Sections A–E form the question paper. Detailed solutions and marking guidance begin only after Question 39; source-pattern and competency rationale follow in a separate appendix.

### Instructions

Attempt all 39 questions. There are no internal choices. Section A: Q1–20, 1 mark each; Section B: Q21–26, 2 each; Section C: Q27–33, 3 each; Section D: Q34–36, 5 each; Section E: Q37–39, 4 each. Show reasoning, substitutions and units in constructed responses.

Unless explicitly stated otherwise, sources maintain their stated terminal voltage; connecting wires and ammeters have zero resistance; voltmeters draw no current. Resistors are ohmic at constant temperature, and resistance changes due to heating are ignored except where a question explicitly investigates them. A meter described as offset has only the indicated additive reading error and otherwise ideal behaviour.

For purely resistive circuits the electrical energy supplied is converted to heat. Do not assume all this heat stays in a measured object unless stated; losses and collection assumptions are given where needed. Figures are schematic, and labelled dimensions/values take precedence

over pictorial scale.

For Q17–20 use these shared options exactly: A: both true and R correctly explains A; B: both true but R does not correctly explain A; C: A true but R false; D: A false but R true.

- Use SI units unless another unit is requested.
- Each figure belongs only to its associated question.
- All numerical datasets and diagrams are original teaching constructions.

## Section A

### Question 1 | 1 mark

special-39-01-q01

A steady current of 75 mA flows for 80 s. What charge crosses one section of the wire? [1 mark]

- A. 0.94 C
- B. 6.0 C
- C. 60 C
- D. 6000 C

### Question 2 | 1 mark

special-39-01-q02

An ideal source supplies 42 J while moving 7 C through its external circuit. Its potential difference is: [1 mark]

- A. 294 V
- B. 35 V
- C. 6 V
- D. 0.167 V

### Question 3 | 1 mark

special-39-01-q03

Which expression has the SI unit of electrical resistivity? [1 mark]

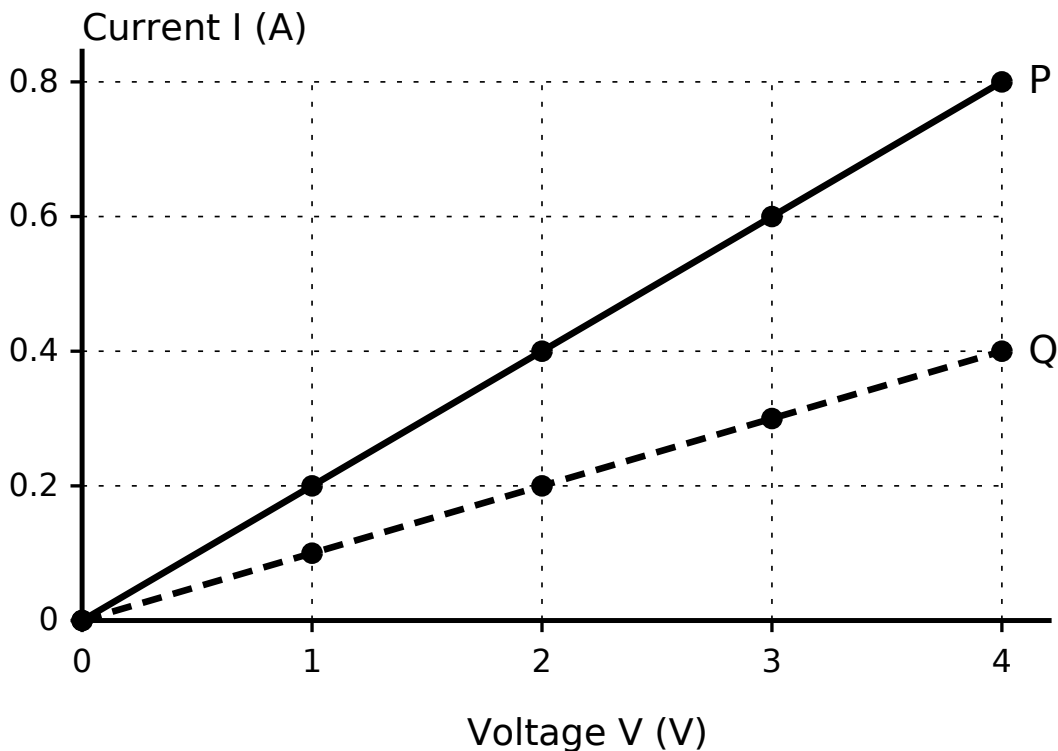
- A.  $\Omega/\text{m}$
- B.  $\Omega \text{ m}^2$
- C.  $\Omega/\text{m}^2$
- D.  $\Omega \text{ m}$

### Question 4 | 1 mark

special-39-01-q04

The supplied graph records two uniform wires P and Q of equal length and cross-sectional area at the same constant temperature. Which conclusion follows? Pay attention to the axes. [1 mark]

## Q4 · Two wire measurements



- A. Q has twice the resistivity of P
- B. P has twice the resistivity of Q
- C. Both have the same resistivity
- D. The graph establishes only their lengths

**Question 5 | 1 mark**

special-39-01-q05

A student obtains  $(V, I) = (1 \text{ V}, 0.10 \text{ A}), (2 \text{ V}, 0.20 \text{ A}), (3 \text{ V}, 0.25 \text{ A})$ . The metal wire is noticeably warmer for the last reading. Which next step best tests whether heating caused the departure? [1 mark]

- A. Erase the third reading
- B. Declare that all metals violate Ohm's law
- C. Cool the wire, then briefly repeat at 3 V
- D. Repeat at 1 V while leaving the wire hot

**Question 6 | 1 mark**

special-39-01-q06

A recorder shows 0.30 A before a resistor and 0.30 A after it in a steady single-loop circuit, while electrical energy is transferred to heat. Which interpretation is correct? [1 mark]

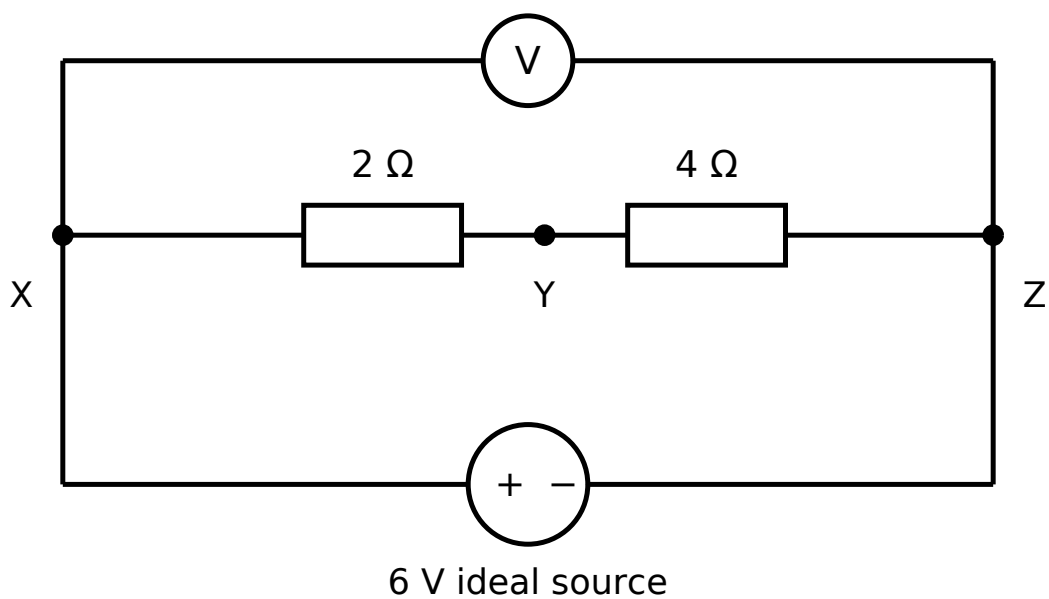
- A. The resistor consumes charge but replaces it with heat

- B. Charge flow continues; electrical energy is transferred to heat
- C. No energy can be transferred because current is unchanged
- D. Current must increase on leaving the resistor

**Question 7 | 1 mark**

special-39-01-q07

In the supplied circuit all meters, wires and the 6 V source are ideal. A student calls the reading of the voltmeter the voltage across the 4  $\Omega$  resistor. Which pair gives (the voltmeter reading, the actual voltage across 4  $\Omega$ )? [1 mark]

**Q7 · Meter lead positions**

Dots mark electrical connections.

- A. (4 V, 6 V)
- B. (2 V, 4 V)
- C. (6 V, 6 V)
- D. (6 V, 4 V)

**Question 8 | 1 mark**

special-39-01-q08

A fixed resistor is tested at the same temperature using an accurate voltmeter. At actual voltages 2 V and 4 V, the ammeter reads 0.24 A and 0.44 A. With no current it reads 0.04 A. Which corrected pair is consistent with Ohm's law? The only meter error is a constant added reading. [1 mark]

- A. (0.20 A, 0.40 A)

B. (0.28 A, 0.48 A)

C. (0.12 A, 0.22 A)

D. (0.24 A, 0.44 A)

### Question 9 | 1 mark

special-39-01-q09

A team wants to compare the resistivities of two different materials from measured resistance. Which pair of samples removes geometric differences without needing a correction? Both are at the same temperature. [1 mark]

A. Equal lengths only

B. Equal masses only

C. Equal lengths and equal cross-sectional areas

D. Equal measured resistances only

### Question 10 | 1 mark

special-39-01-q10

A uniform wire has resistance  $R$ . Another wire of the same material at the same temperature has the same length and twice the diameter. Its resistance is: [1 mark]

A.  $R/2$

B.  $R/4$

C.  $2R$

D.  $4R$

### Question 11 | 1 mark

special-39-01-q11

Two positive resistors,  $9\ \Omega$  and  $18\ \Omega$ , are said to be connected in parallel. A report lists their equivalent resistance as  $12\ \Omega$ . Which criticism is valid without a full calculation? [1 mark]

A. Parallel resistance must exceed  $18\ \Omega$

B. Parallel resistance must equal  $27\ \Omega$

C.  $12\ \Omega$  is valid because it lies between the branch values

D. Parallel resistance must be below  $9\ \Omega$

### Question 12 | 1 mark

special-39-01-q12

Two heater trials use the same constant-resistance coil. In trial Y the current is half that in trial X but the duration is four times as long. Compare the electrical heat generated, ignoring energy losses from the coil. [1 mark]

A. Equal in X and Y

- B. Four times greater in Y
- C. Twice as great in Y
- D. Half as great in Y

**Question 13 | 1 mark**

special-39-01-q13

A 20 W laboratory lamp operates continuously for 3 h at its rated power. Its electrical energy use is: [1 mark]

- A. 60 kWh
- B. 6 kWh
- C. 0.060 kWh
- D. 0.0060 kWh

**Question 14 | 1 mark**

special-39-01-q14

Two constant-resistance coils X and Y have  $R_X > R_Y$ . They are each connected directly to the same ideal fixed-voltage source in separate trials of equal duration. A student predicts more heat in X because  $H = I^2 R t$ . What is the correct decision? [1 mark]

- A. Accept: larger R always means more heat
- B. Reject: X generates less because its current is smaller
- C. Accept: both currents must be equal
- D. Reject: X generates no heat

**Question 15 | 1 mark**

special-39-01-q15

A resistor dissipates 3 W when connected to a 6 V source. A second resistor of the same resistance is added in parallel across that ideal source. All resistances remain constant. Which observation should stay unchanged? [1 mark]

- A. Total source current
- B. Total source power
- C. Equivalent resistance
- D. Power in the original resistor

**Question 16 | 1 mark**

special-39-01-q16

Which statement correctly defines one watt? [1 mark]

- A. One joule transferred each second
- B. One joule transferred each coulomb

C. One coulomb transferred each second

D. One joule transferred each hour

**Question 17 | 1 mark**

special-39-01-q17

Assertion (A): A straight recorded I-V line that misses the origin does not, by itself, establish failure of Ohm's law for the resistor.

Reason (R): A constant ammeter zero offset can shift every recorded current by the same amount.

Choose the shared assertion-reason option. [1 mark]

A. both true and R correctly explains A

B. both true but R does not correctly explain A

C. A true but R false

D. A false but R true

**Question 18 | 1 mark**

special-39-01-q18

Assertion (A): A student may establish equal resistivity for two wires merely by observing equal resistance at the same temperature.

Reason (R): Resistance also depends on the length-to-area ratio of a wire.

Choose the shared assertion-reason option. [1 mark]

A. both true and R correctly explains A

B. both true but R does not correctly explain A

C. A true but R false

D. A false but R true

**Question 19 | 1 mark**

special-39-01-q19

Assertion (A): Two ideal resistors connected in series carry the same steady current.

Reason (R): Two unequal resistors connected in parallel across an ideal source have equal potential differences.

Choose the shared assertion-reason option. [1 mark]

A. both true and R correctly explains A

B. both true but R does not correctly explain A

C. A true but R false

D. A false but R true

**Question 20 | 1 mark**

special-39-01-q20

Assertion (A): A kilowatt hour is a unit of electrical energy.

Reason (R): A kilowatt hour is equal to 1000 joules.

Choose the shared assertion-reason option. [1 mark]

- A. both true and R correctly explains A
- B. both true but R does not correctly explain A
- C. A true but R false
- D. A false but R true

## Section B

### Question 21 | 2 marks

special-39-01-q21

A charge counter records 9 C crossing a wire in the first 30 s and a further 9 C in the next 10 s. (i) Find the average current in each interval. [1 mark] (ii) A student says that equal charges mean equal currents. Evaluate the claim using these observations. [1 mark]

### Question 22 | 2 marks

special-39-01-q22

Two energy-transfer records for the same ideal source are ( $Q = 4 \text{ C}$ ,  $W = 18 \text{ J}$ ) and ( $Q = 10 \text{ C}$ ,  $W = 45 \text{ J}$ ). (i) Does the second record show that the source voltage increased? Justify quantitatively. [1 mark] (ii) Predict the work for a further 6 C at the same voltage and explain why work, unlike voltage, depends on the amount of charge. [1 mark]

### Question 23 | 2 marks

special-39-01-q23

A laboratory has two analogue ammeters: L has range 0–0.30 A with 30 equal divisions across its full scale; H has range 0–3.0 A with 30 equal divisions across its full scale. Both have negligible resistance, correct zero and no calibration error. Each is connected in series when used. For this task, a current change is considered resolvable only if it moves the pointer by at least one whole division; do not estimate fractions of a division. A chosen meter must also accommodate every expected current without exceeding full scale. (i) In the first test, current will stay between 0.18 A and 0.24 A, and changes of 0.02 A must be resolved. Choose L or H and justify the choice using both full scale and division size. [1 mark] (ii) In a second test, the current may reach 0.45 A, while the same 0.02 A changes must still be resolved. A student proposes switching to H. Does either available meter now meet both requirements? Justify your decision and state the full-scale and division-size requirements for a suitable replacement. [1 mark]

### Question 24 | 2 marks

special-39-01-q24

A uniform wire has length 0.50 m, cross-sectional area  $0.20 \text{ mm}^2$  and measured resistance  $2.0 \Omega$  at a fixed temperature. (i) Calculate its resistivity in SI units. [1 mark] (ii) The wire is cut into two equal lengths without stretching. A student says each half has half the resistivity. Correct the claim and state each half's resistance. [1 mark]

### Question 25 | 2 marks

special-39-01-q25

A  $2 \Omega$  and a  $5 \Omega$  resistor are in series across an ideal 7 V source. An accurate ammeter in the single loop reads 1 A. A voltmeter placed in turn across each resistor reports 3 V and 6 V. All resistances and temperature remain constant. (i) Show why the two voltage readings cannot both be correct. [1 mark] (ii) If the voltmeter has one constant added offset and no other error, determine it and state the corrected readings. [1 mark]

### Question 26 | 2 marks

special-39-01-q26

Two unlabelled resistors are connected in parallel across an ideal 6 V source. Accurate ammeters show 0.75 A from the source and 0.25 A in one branch. (i) Infer the current in the other branch and explain the inference. [1 mark] (ii) A label claiming that the other branch resistor is  $24\ \Omega$  is proposed. Accept or reject it quantitatively. [1 mark]

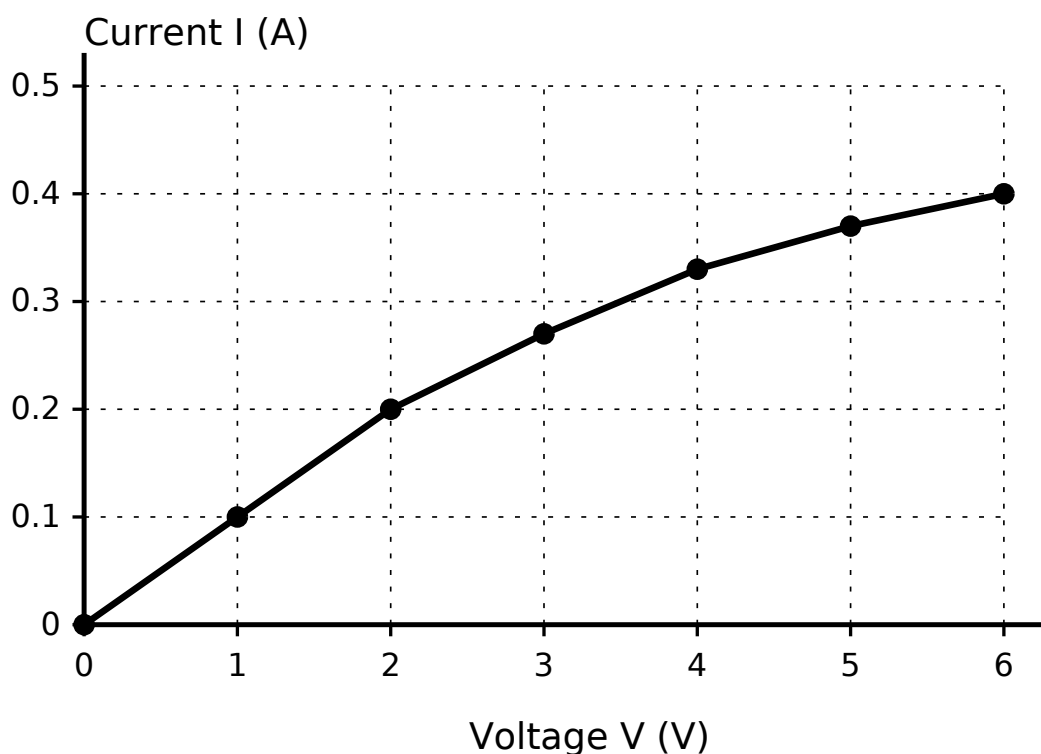
## Section C

### Question 27 | 3 marks

special-39-01-q27

The supplied plot shows an accurately measured I-V trace for a metal wire during a slow voltage sweep; the wire was not temperature-controlled. (i) Read the currents at 2 V and 6 V and compare the corresponding  $V/I$  values. [1 mark] (ii) Does the plotted evidence support constant resistance over the whole range? Explain. [1 mark] (iii) Propose a repeat procedure that tests heating as an explanation without assuming that the plot alone proves it. [1 mark]

#### Q27 · Slow voltage sweep

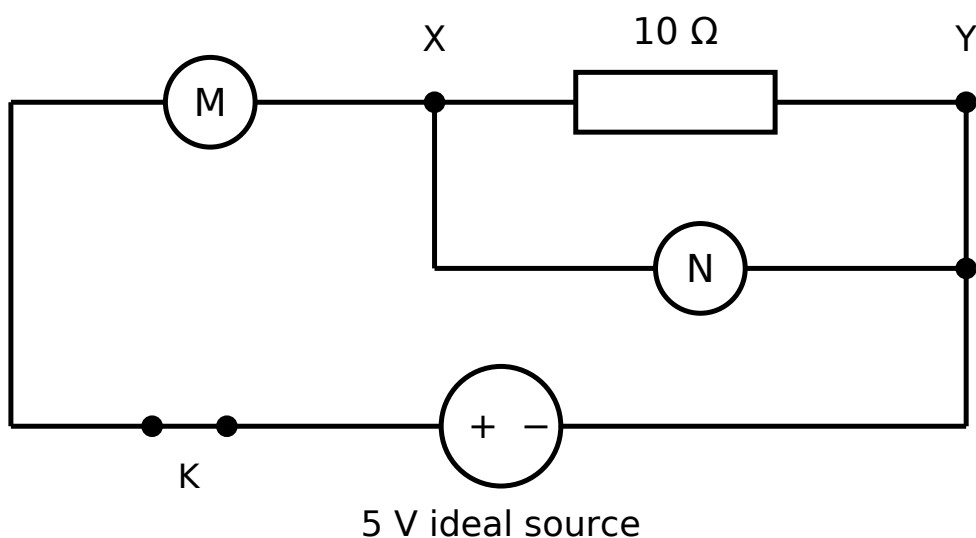


### Question 28 | 3 marks

special-39-01-q28

The supplied circuit is a proposed resistance-measurement setup. The switch is closed, the 5 V source is ideal, meter M is an ideal voltmeter, and meter N is an ideal ammeter. The resistor is  $10\ \Omega$ . (i) Explain why M is incorrectly placed and predict the circuit current. [1 mark] (ii) Explain why N is incorrectly placed; also explain why N does not cause an unbounded source current in this particular faulty setup. [1 mark] (iii) State the rewiring needed for a valid measurement and give the two corrected readings. Disconnect the source before rewiring. [1 mark]

## Q28 • Proposed measurement circuit



M: voltmeter N: ammeter Switch closed

## Question 29 | 3 marks

special-39-01-q29

A  $6\ \Omega$  and a  $12\ \Omega$  resistor are joined in parallel across an ideal  $6\ \text{V}$  source. The  $12\ \Omega$  branch is then disconnected, leaving the  $6\ \Omega$  branch intact. (i) Calculate the initial equivalent resistance. [1 mark] (ii) Predict the change in total source current and explain why it changes. [1 mark] (iii) A student says the remaining branch now receives the old total current. Test that claim with its new current and voltage. [1 mark]

## Question 30 | 3 marks

special-39-01-q30

A team compares electrical heat generated in two identical constant-resistance coils. Trial A uses  $0.40\ \text{A}$  for  $150\ \text{s}$ . Trial B uses  $0.60\ \text{A}$  for  $100\ \text{s}$ . No heat escapes the collecting system, and the heat generated equals the measured collected energy. (i) Without needing the coil resistance, find  $H_B/H_A$ . [1 mark] (ii) A student argues that the heat must be equal because the same charge passed. Evaluate this argument, including the charge values. [1 mark] (iii) Find the duration at  $0.60\ \text{A}$  that would generate the same heat as trial A. [1 mark]

## Question 31 | 3 marks

special-39-01-q31

A resistive test lamp is labelled  $12\ \text{V}$ ,  $6\ \text{W}$ . Its resistance is assumed constant at the rated value. (i) Determine that resistance. [1 mark] (ii) A technician proposes a  $6\ \text{V}$  source, arguing that the lamp will then use  $3\ \text{W}$  because the voltage is halved. Test the claim and state the actual power. [1 mark] (iii) Can your result alone predict the exact perceived brightness of a real filament lamp whose resistance changes with temperature? Explain. [1 mark]

**Question 32 | 3 marks**

special-39-01-q32

A laboratory energy log lists a 40 W inspection lamp on continuously for 90 min and a 200 W heater on for only 18 min during the same session. These are the only devices used and their actual powers remain as stated. A recorder writes the session total as 0.36 kWh. (i) Calculate the expected total use in kWh. [1 mark] (ii) Is the recorded total consistent with the operating times? Support your conclusion with the expected value. [1 mark] (iii) The recorder counted the heater as on for the full session. Show whether this timing error explains the recorded total exactly; do not assume an electrical fault. [1 mark]

**Question 33 | 3 marks**

special-39-01-q33

Two sealed boxes each contain a single uniform wire. At the same controlled temperature, both have measured resistance  $8\ \Omega$ . Box X contains a wire of length  $L$  and area  $A$ . Box Y contains one of length  $2L$  and area  $3A$ . (i) Derive  $\rho_Y/\rho_X$  from these measurements. [1 mark] (ii) A student identifies both wires as the same material solely because their resistances agree. Is the identification supported? [1 mark] (iii) Suggest a different geometry for a new sample of material Y that would allow a direct resistance comparison with X to compare resistivities. [1 mark]

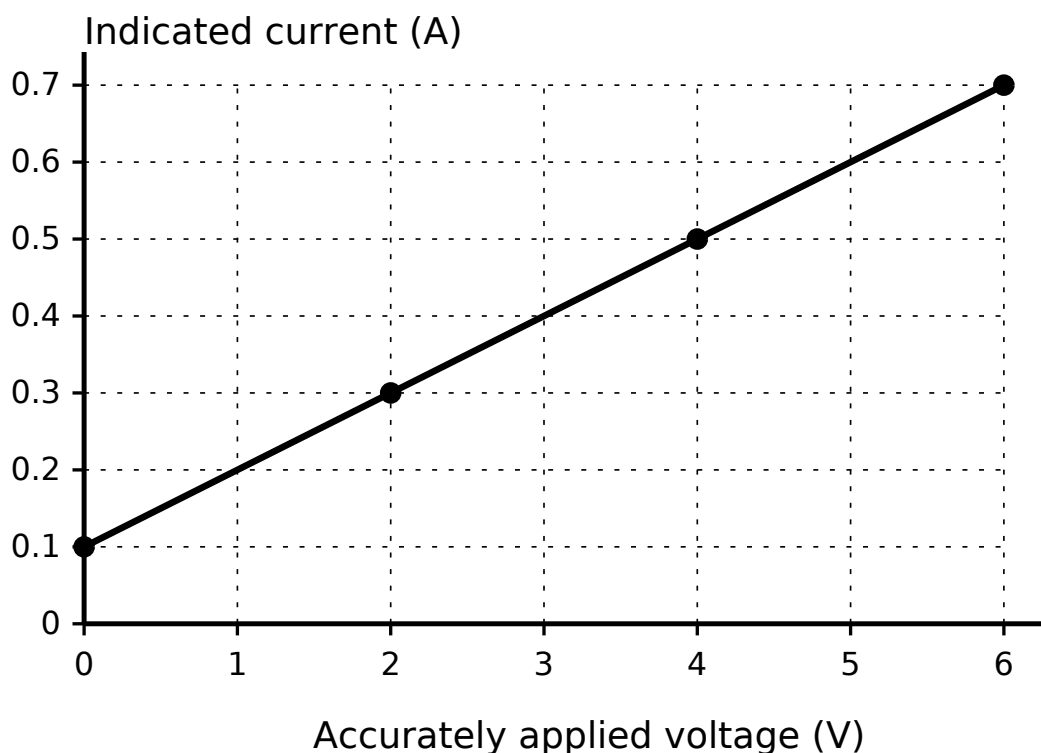
## Section D

### Question 34 | 5 marks

special-39-01-q34

A laboratory ammeter is checked using one ohmic resistor kept at constant temperature. The applied voltage is measured accurately, including a no-current check at 0 V. The supplied graph contains the observations. The ammeter has negligible resistance and its only possible fault is a constant added current indication. (i) Read the no-current indication and state the correction to every reading. [1 mark] (ii) Use the change between the plotted 2 V and 6 V observations to determine the resistor's resistance. Explain in ordinary words why subtracting the two indications removes a constant offset. Do not use statistical fitting. [2 marks] (iii) A student divides 2 V by the indicated current at 2 V and reports the result as the resistance. Calculate that result and explain the discrepancy. [1 mark] (iv) Predict the ammeter indication at an accurately applied 3 V after leaving the offset unadjusted. [1 mark]

#### Q34 · Ammeter calibration observations



### Question 35 | 5 marks

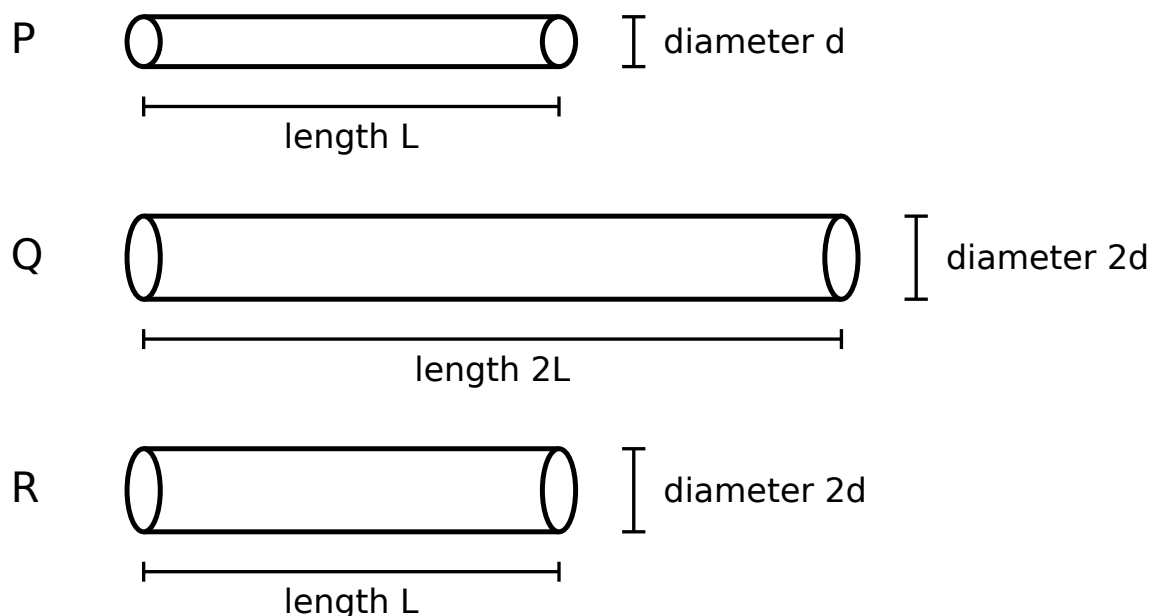
special-39-01-q35

A team is given the three uniform cylindrical wire specimens in the supplied diagram. All are made from the same alloy and are held at the same temperature;  $L$  and  $d$  are positive. Each specimen is tested separately at an accurately measured 1.2 V, using ideal meters and negligible-resistance leads. (i) Select the pair that isolates the effect of length, and justify your choice from the figure. [1 mark] (ii) Select the pair that isolates the effect of diameter, and state the ratio of their cross-sectional areas. [1 mark] (iii) The current for P is 0.12 A. Predict the currents for Q and R, using the displayed geometries. [2 marks] (iv) A team instead compares only P with Q and concludes that doubling length halves resistance. Explain why this is not a

valid length-only conclusion even if its measured resistance ratio is correct. [1 mark]

### Q35 • Wire specimen geometry

Same alloy and temperature • not to scale



### Question 36 | 5 marks

special-39-01-q36

A technician is checking a  $10\ \Omega$  standard resistor using an ideal  $6\ \text{V}$  source and accurate, ideal meters. The ammeter is in the single series path; the voltmeter is connected directly to the standard resistor's own terminals. The source terminal voltage is confirmed as  $6\ \text{V}$ , but the readings are  $I = 0.40\ \text{A}$  and  $V_{\text{resistor}} = 4.0\ \text{V}$ . Exactly one fault is present: either (A) the standard resistor has changed resistance, or (B) one contact elsewhere in the series path has acquired resistance. All other leads and contacts have zero resistance. (i) Use the two meter readings to decide whether the standard itself is still  $10\ \Omega$ . [1 mark] (ii) Choose A or B and determine the faulty contact resistance. [2 marks] (iii) Specify a voltage measurement that would confirm the location of the fault without moving the ammeter, and predict its reading. [1 mark] (iv) Calculate the electrical power generated as heat in the faulty contact. [1 mark]

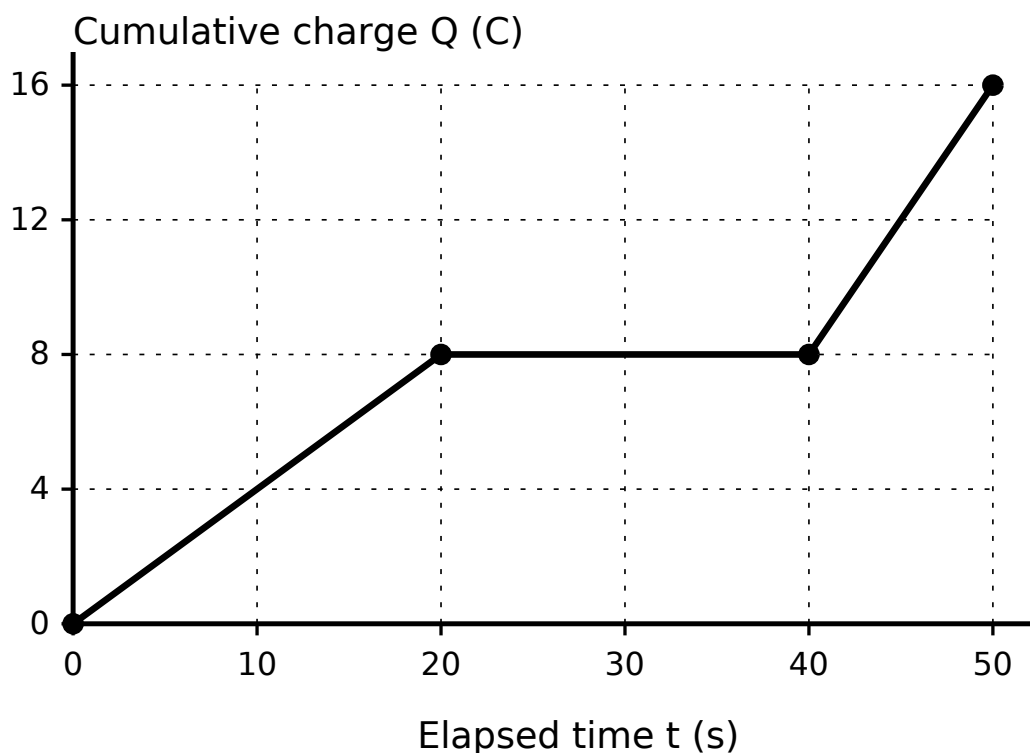
## Section E

### Question 37 | 4 marks

special-39-01-q37

A charge counter records cumulative charge crossing a section of a test circuit. Its readings form the supplied piecewise straight graph; segments represent constant current, not instantaneous jumps in charge. The counter is in the source lead and records all charge delivered by an ideal 3 V source. A switch selects different fixed resistive loads for the two rising intervals and opens the circuit during the horizontal interval. Ignore switching transients; each selected load has constant resistance. (i) Identify the interval in which the circuit carries no current and justify this from the graph. [1 mark] (ii) Determine the currents in the first and final rising intervals, using changes in charge and time. [1 mark] (iii) A student claims that the equal charge increments in those two intervals require equal power. Test the claim by comparing their electrical powers and total energies. [2 marks]

#### Q37 · Cumulative charge record



### Question 38 | 4 marks

special-39-01-q38

A lab group records the following accurate observations for two constant-resistance components, keeping each at the same temperature throughout: X alone: 3 V, 0.30 A; Y alone: 3 V, 0.15 A. The components are then put inside a sealed box using either series or parallel connection, with no other components and with zero-resistance connecting leads. The resistors dissipate electrical energy as heat. Across the box the group measures 6 V and 0.20 A. (i) Find  $R_X$  and  $R_Y$  from the individual observations. [1 mark] (ii) Identify the connection inside the box using the box measurements. [1 mark] (iii) Another group claims the box could be parallel if the current merely divided equally. Test that alternative by calculating its total current at 6 V and

explaining why equal division is not valid here. [2 marks]

**Question 39 | 4 marks**

special-39-01-q39

A heating experiment uses a constant-resistance coil at an accurately maintained 5 V and 0.40 A for 120 s. A heat collector captures only the energy reaching it from the coil; it reports 180 J. Any remaining electrical energy is transferred as heat to the surroundings, and there is no other energy input. A student writes: “The coil resistance must be wrong because the collector should read all the electrical energy.” (i) Find the electrical energy supplied and the energy not captured by the collector. [1 mark] (ii) Does the collector reading alone support the student’s diagnosis? Use the electrical readings and the stated energy pathway. [1 mark] (iii) In a repeat, voltage, current and duration are unchanged but improved insulation gives a collector reading of 225 J. Decide whether the coil generated more electrical heat, and quantify what changed. [2 marks]

## Detailed solutions and marking guidance

Original model answers, numbered to match the question paper. These are not an official marking scheme.

### Solution 1 — special-39-01-q01

Correct option: B

Correct option: B.

[1 mark]  $Q = It = 0.075 \times 80 = 6.0 \text{ C}$ . Convert milliamperes to amperes before multiplication.

#### Mark allocation

$Q = It = 0.075 \times 80 = 6.0 \text{ C}$ . Convert milliamperes to amperes before multiplication. [1 marks]

### Solution 2 — special-39-01-q02

Correct option: C

Correct option: C.

[1 mark]  $V = W/Q = 42/7 = 6 \text{ V}$ ; potential difference is energy transferred per coulomb.

#### Mark allocation

$V = W/Q = 42/7 = 6 \text{ V}$ ; potential difference is energy transferred per coulomb. [1 marks]

### Solution 3 — special-39-01-q03

Correct option: D

Correct option: D.

[1 mark] From  $\rho = RA/L$ , the unit is  $\Omega \times \text{m}^2/\text{m} = \Omega \text{ m}$ .

#### Mark allocation

From  $\rho = RA/L$ , the unit is  $\Omega \times \text{m}^2/\text{m} = \Omega \text{ m}$ . [1 marks]

### Solution 4 — special-39-01-q04

Correct option: A

Correct option: A.

[1 mark] A: at 4 V the graph gives  $I_P = 0.8 \text{ A}$  and  $I_Q = 0.4 \text{ A}$ . Thus  $R_P = 5 \Omega$  and  $R_Q = 10 \Omega$ . Equal geometry means  $\rho_Q/\rho_P = R_Q/R_P = 2$ . The steeper I-versus-V line has lower resistance.

#### Mark allocation

A: at 4 V the graph gives  $I_P = 0.8 \text{ A}$  and  $I_Q = 0.4 \text{ A}$ . Thus  $R_P = 5 \Omega$  and  $R_Q = 10 \Omega$ . Equal geometry means  $\rho_Q/\rho_P = R_Q/R_P = 2$ . The steeper I-versus-V line has lower resistance. [1 marks]

### Solution 5 — special-39-01-q05

Correct option: C

Correct option: C.

[1 mark] Cool the wire back to the initial temperature and repeat the 3 V reading briefly. This isolates temperature rather than changing voltage and temperature together; a repeat near 0.30 A would support the heating explanation, not prove it uniquely.

**Mark allocation**

Cool the wire back to the initial temperature and repeat the 3 V reading briefly. This isolates temperature rather than changing voltage and temperature together; a repeat near 0.30 A would support the heating explanation, not prove it uniquely. [1 marks]

**Solution 6 — special-39-01-q06**

Correct option: B

Correct option: B.

[1 mark] The equal currents show no steady accumulation or consumption of charge in the resistor. Electrical energy is transferred into heat; current is not used up.

**Mark allocation**

The equal currents show no steady accumulation or consumption of charge in the resistor. Electrical energy is transferred into heat; current is not used up. [1 marks]

**Solution 7 — special-39-01-q07**

Correct option: D

Correct option: D.

[1 mark] The voltmeter leads span both resistors, so it reads 6 V. The series current is  $6/(2 + 4) = 1$  A; hence the 4  $\Omega$  resistor has 4 V across it. The correct pair is (6 V, 4 V).

**Mark allocation**

The voltmeter leads span both resistors, so it reads 6 V. The series current is  $6/(2 + 4) = 1$  A; hence the 4  $\Omega$  resistor has 4 V across it. The correct pair is (6 V, 4 V). [1 marks]

**Solution 8 — special-39-01-q08**

Correct option: A

Correct option: A.

[1 mark] Subtract the no-current reading from each indication: 0.20 A and 0.40 A. Then  $2/0.20 = 4/0.40 = 10 \Omega$ . An additive error must be subtracted, not scaled.

**Mark allocation**

Subtract the no-current reading from each indication: 0.20 A and 0.40 A. Then  $2/0.20 = 4/0.40 = 10 \Omega$ . An additive error must be subtracted, not scaled. [1 marks]

**Solution 9 — special-39-01-q09**

Correct option: C

Correct option: C.

[1 mark] Equal lengths and equal cross-sectional areas give  $R = \rho L/A$  with the same  $L/A$ . A difference in  $R$  can then be attributed to a difference in  $\rho$  within the stated controls.

**Mark allocation**

Equal lengths and equal cross-sectional areas give  $R = \rho L/A$  with the same  $L/A$ . A difference in  $R$  can then be attributed to a difference in  $\rho$  within the stated controls. [1 marks]

**Solution 10 — special-39-01-q10**

Correct option: B

Correct option: B.

[1 mark] Area is proportional to diameter squared. Doubling diameter makes area four times as large, so resistance becomes  $R/4$ .

**Mark allocation**

Area is proportional to diameter squared. Doubling diameter makes area four times as large, so resistance becomes  $R/4$ . [1 marks]

**Solution 11 — special-39-01-q11**

Correct option: D

Correct option: D.

[1 mark] The parallel combination must have resistance less than the smaller branch resistance,  $9\ \Omega$ , because it provides an additional conducting path. Thus  $12\ \Omega$  is impossible; the actual value would be  $6\ \Omega$ .

**Mark allocation**

The parallel combination must have resistance less than the smaller branch resistance,  $9\ \Omega$ , because it provides an additional conducting path. Thus  $12\ \Omega$  is impossible; the actual value would be  $6\ \Omega$ . [1 marks]

**Solution 12 — special-39-01-q12**

Correct option: A

Correct option: A.

[1 mark]  $H_Y/H_X = (I_Y/I_X)^2(t_Y/t_X) = (1/2)^2 \times 4 = 1$ , since the same  $R$  applies. The heat generated is equal.

**Mark allocation**

$H_Y/H_X = (I_Y/I_X)^2(t_Y/t_X) = (1/2)^2 \times 4 = 1$ , since the same  $R$  applies. The heat generated is equal. [1 marks]

**Solution 13 — special-39-01-q13**

Correct option: C

Correct option: C.

[1 mark]  $E = Pt = 20\ \text{W} \times 3\ \text{h} = 60\ \text{Wh} = 0.060\ \text{kWh}$ . Power in watts alone is not energy.

**Mark allocation**

$E = Pt = 20\ \text{W} \times 3\ \text{h} = 60\ \text{Wh} = 0.060\ \text{kWh}$ . Power in watts alone is not energy. [1 marks]

**Solution 14 — special-39-01-q14**

Correct option: B

Correct option: B.

[1 mark] Reject the prediction: the currents are not equal. Substituting  $I = V/R$  gives  $H = V^2t/R$ , so the higher-resistance coil  $X$  generates less heat.

**Mark allocation**

Reject the prediction: the currents are not equal. Substituting  $I = V/R$  gives  $H = V^2t/R$ , so the higher-resistance coil X generates less heat. [1 marks]

**Solution 15 — special-39-01-q15**

Correct option: D

Correct option: D.

[1 mark] The original resistor still has 6 V across it, so its current and its 3 W power remain unchanged. The total current and total power increase, and equivalent resistance decreases.

**Mark allocation**

The original resistor still has 6 V across it, so its current and its 3 W power remain unchanged. The total current and total power increase, and equivalent resistance decreases. [1 marks]

**Solution 16 — special-39-01-q16**

Correct option: A

Correct option: A.

[1 mark] One watt is a rate of energy transfer of one joule per second. One joule is energy, while one coulomb per second defines one ampere.

**Mark allocation**

One watt is a rate of energy transfer of one joule per second. One joule is energy, while one coulomb per second defines one ampere. [1 marks]

**Solution 17 — special-39-01-q17**

Correct option: A

Correct option: A.

[1 mark] Both statements are true, and R explains A. Even if true  $I = V/R$ , adding a fixed meter offset shifts the plotted line away from the origin. The meter must be checked before blaming the resistor.

**Mark allocation**

Both statements are true, and R explains A. Even if true  $I = V/R$ , adding a fixed meter offset shifts the plotted line away from the origin. The meter must be checked before blaming the resistor. [1 marks]

**Solution 18 — special-39-01-q18**

Correct option: D

Correct option: D.

[1 mark] A is false but R is true. Since  $R = \rho L/A$ , equal R can result from unequal  $\rho$  with compensating geometry. Geometry information is needed.

**Mark allocation**

A is false but R is true. Since  $R = \rho L/A$ , equal R can result from unequal  $\rho$  with compensating geometry. Geometry information is needed. [1 marks]

**Solution 19 — special-39-01-q19**

Correct option: B

Correct option: B.

[1 mark] Both are true, but R does not explain A: it describes a different connection. Series current is the same because there is a single steady path with no charge accumulation.

**Mark allocation**

Both are true, but R does not explain A: it describes a different connection. Series current is the same because there is a single steady path with no charge accumulation. [1 marks]

**Solution 20 — special-39-01-q20**

Correct option: C

Correct option: C.

[1 mark] A is true, but R is false:  $1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$ , not  $1000 \text{ J}$ .

**Mark allocation**

A is true, but R is false:  $1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$ , not  $1000 \text{ J}$ . [1 marks]

**Solution 21 — special-39-01-q21**

[1 mark] First interval:  $I_1 = 9/30 = 0.30 \text{ A}$ . Second:  $I_2 = 9/10 = 0.90 \text{ A}$ . Award 0.5 for each.

[1 mark] The claim is false: current is charge per unit time. The second interval transfers the same charge in one-third the time, so its average current is three times as large.

**Mark allocation**

First interval:  $I_1 = 9/30 = 0.30 \text{ A}$ . Second:  $I_2 = 9/10 = 0.90 \text{ A}$ . Award 0.5 for each. [1 marks]

The claim is false: current is charge per unit time. The second interval transfers the same charge in one-third the time, so its average current is three times as large. [1 marks]

**Solution 22 — special-39-01-q22**

[1 mark] No.  $W/Q = 18/4 = 4.5/10 = 4.5 \text{ J/C}$ , so both records show the same  $4.5 \text{ V}$ .

[1 mark]  $W = VQ = 4.5 \times 6 = 27 \text{ J}$ . Each coulomb transfers  $4.5 \text{ J}$ , so more coulombs mean more total work while the energy per coulomb stays fixed.

**Mark allocation**

No.  $W/Q = 18/4 = 4.5/10 = 4.5 \text{ J/C}$ , so both records show the same  $4.5 \text{ V}$ . [1 marks]

$W = VQ = 4.5 \times 6 = 27 \text{ J}$ . Each coulomb transfers  $4.5 \text{ J}$ , so more coulombs mean more total work while the energy per coulomb stays fixed. [1 marks]

**Solution 23 — special-39-01-q23**

[1 mark] Choose L. Its division size is  $0.30/30 = 0.010 \text{ A}$ , so a  $0.02 \text{ A}$  change spans two divisions, and the largest expected current,  $0.24 \text{ A}$ , is below its  $0.30 \text{ A}$  full scale. H can accommodate the current but has  $3.0/30 = 0.10 \text{ A}$  per division; a  $0.02 \text{ A}$  change spans only 0.2 division and fails the stated resolution rule. Award the mark for the choice justified against both range and resolution, not division-size arithmetic alone.

[1 mark] Neither meter meets both requirements in the second test. L would be over range because  $0.45 \text{ A}$  exceeds  $0.30 \text{ A}$ . H has enough range but still cannot resolve a  $0.02 \text{ A}$  change

under the one-whole-division rule. Use a replacement with full scale at least 0.45 A and division size no greater than 0.02 A; for example, a 0–0.60 A meter with 60 equal divisions has 0.010 A per division and satisfies both. Award the mark for rejecting both available meters on the correct grounds and specifying both replacement requirements.

### Mark allocation

Choose L. Its division size is  $0.30/30 = 0.010$  A, so a 0.02 A change spans two divisions, and the largest expected current, 0.24 A, is below its 0.30 A full scale. H can accommodate the current but has  $3.0/30 = 0.10$  A per division; a 0.02 A change spans only 0.2 division and fails the stated resolution rule. Award the mark for the choice justified against both range and resolution, not division-size arithmetic alone. [1 marks]

Neither meter meets both requirements in the second test. L would be over range because 0.45 A exceeds 0.30 A. H has enough range but still cannot resolve a 0.02 A change under the one-whole-division rule. Use a replacement with full scale at least 0.45 A and division size no greater than 0.02 A; for example, a 0–0.60 A meter with 60 equal divisions has 0.010 A per division and satisfies both. Award the mark for rejecting both available meters on the correct grounds and specifying both replacement requirements. [1 marks]

### Solution 24 — special-39-01-q24

[1 mark]  $A = 0.20 \times 10^{-6} \text{ m}^2 = 2.0 \times 10^{-7} \text{ m}^2$ .  $\rho = RA/L = 2.0 \times 2.0 \times 10^{-7}/0.50 = 8.0 \times 10^{-7} \Omega \text{ m}$ .

[1 mark] Each half has resistance  $1.0 \Omega$  because length halves while area is unchanged. Resistivity stays  $8.0 \times 10^{-7} \Omega \text{ m}$  because material and temperature have not changed.

### Mark allocation

$A = 0.20 \times 10^{-6} \text{ m}^2 = 2.0 \times 10^{-7} \text{ m}^2$ .  $\rho = RA/L = 2.0 \times 2.0 \times 10^{-7}/0.50 = 8.0 \times 10^{-7} \Omega \text{ m}$ . [1 marks]

Each half has resistance  $1.0 \Omega$  because length halves while area is unchanged. Resistivity stays  $8.0 \times 10^{-7} \Omega \text{ m}$  because material and temperature have not changed. [1 marks]

### Solution 25 — special-39-01-q25

[1 mark] The drops should add to 7 V, but  $3 + 6 = 9$  V. Independently, IR predicts 2 V and 5 V at 1 A.

[1 mark] Each reading is 1 V too high:  $3 - 2 = 6 - 5 = 1$  V. The offset is +1 V; subtract 1 V to obtain 2 V and 5 V. The total excess of 2 V is shared over two readings.

### Mark allocation

The drops should add to 7 V, but  $3 + 6 = 9$  V. Independently, IR predicts 2 V and 5 V at 1 A. [1 marks]

Each reading is 1 V too high:  $3 - 2 = 6 - 5 = 1$  V. The offset is +1 V; subtract 1 V to obtain 2 V and 5 V. The total excess of 2 V is shared over two readings. [1 marks]

### Solution 26 — special-39-01-q26

[1 mark] Other branch current =  $0.75 - 0.25 = 0.50$  A, since the source current equals the sum of the steady branch currents.

[1 mark] Reject: both branches have 6 V across them; the other branch resistance is  $6/0.50 = 12 \Omega$ , not  $24 \Omega$ . A  $24 \Omega$  branch would carry only 0.25 A.

**Mark allocation**

Other branch current =  $0.75 - 0.25 = 0.50$  A, since the source current equals the sum of the steady branch currents. [1 marks]

Reject: both branches have 6 V across them; the other branch resistance is  $6/0.50 = 12\ \Omega$ , not  $24\ \Omega$ . A  $24\ \Omega$  branch would carry only 0.25 A. [1 marks]

**Solution 27 — special-39-01-q27**

[1 mark] At 2 V,  $I = 0.20$  A and  $V/I = 10\ \Omega$ . At 6 V,  $I = 0.40$  A and  $V/I = 15\ \Omega$ . Award 0.5 for each correctly read and calculated pair.

[1 mark] No. The inferred resistance rises from  $10\ \Omega$  to  $15\ \Omega$  rather than remaining constant; the points are not on one straight line through the origin. This tests constant  $R$  under these conditions, not Ohm's law at a controlled temperature.

[1 mark] Let the wire return to the same initial temperature before each setting, close the switch only briefly, and record promptly. Compare the repeated high-voltage  $V/I$  with the low-voltage value. Restoration of the same ratio supports heating as the cause; persistent deviation calls for further investigation.

**Mark allocation**

At 2 V,  $I = 0.20$  A and  $V/I = 10\ \Omega$ . At 6 V,  $I = 0.40$  A and  $V/I = 15\ \Omega$ . Award 0.5 for each correctly read and calculated pair. [1 marks]

No. The inferred resistance rises from  $10\ \Omega$  to  $15\ \Omega$  rather than remaining constant; the points are not on one straight line through the origin. This tests constant  $R$  under these conditions, not Ohm's law at a controlled temperature. [1 marks]

Let the wire return to the same initial temperature before each setting, close the switch only briefly, and record promptly. Compare the repeated high-voltage  $V/I$  with the low-voltage value. Restoration of the same ratio supports heating as the cause; persistent deviation calls for further investigation. [1 marks]

**Solution 28 — special-39-01-q28**

[1 mark] M is in the only source path. An ideal voltmeter draws no current, so it behaves as an open path in series and the source current is zero.

[1 mark] N is across the  $10\ \Omega$  resistor; its zero resistance bypasses the resistor. Nevertheless M already breaks the source path, so no complete zero-resistance path spans the source and no unbounded source current occurs.

[1 mark] Put N in series with the resistor and M across the resistor endpoints X and Y. With the ideal 5 V source the readings are  $I_N = 5/10 = 0.50$  A and  $V_M = 5$  V. The corrected ammeter must not remain across X and Y.

**Mark allocation**

M is in the only source path. An ideal voltmeter draws no current, so it behaves as an open path in series and the source current is zero. [1 marks]

N is across the  $10\ \Omega$  resistor; its zero resistance bypasses the resistor. Nevertheless M already breaks the source path, so no complete zero-resistance path spans the source and no unbounded source current occurs. [1 marks]

Put N in series with the resistor and M across the resistor endpoints X and Y. With the ideal 5 V source the readings are  $I_N = 5/10 = 0.50$  A and  $V_M = 5$  V. The corrected ammeter must not remain across X and Y. [1 marks]

### Solution 29 — special-39-01-q29

[1 mark]  $1/R_{eq} = 1/6 + 1/12 = 1/4$ , so  $R_{eq} = 4 \Omega$ .

[1 mark] Initially the source current is  $6/4 = 1.5$  A. After removal it is  $6/6 = 1.0$  A; the lost branch no longer contributes its 0.5 A.

[1 mark] The remaining branch still has 6 V and  $6 \Omega$ , so its current remains 1.0 A, not 1.5 A. An ideal voltage source does not maintain the old total current.

### Mark allocation

$1/R_{eq} = 1/6 + 1/12 = 1/4$ , so  $R_{eq} = 4 \Omega$ . [1 marks]

Initially the source current is  $6/4 = 1.5$  A. After removal it is  $6/6 = 1.0$  A; the lost branch no longer contributes its 0.5 A. [1 marks]

The remaining branch still has 6 V and  $6 \Omega$ , so its current remains 1.0 A, not 1.5 A. An ideal voltage source does not maintain the old total current. [1 marks]

### Solution 30 — special-39-01-q30

[1 mark]  $H_B/H_A = (0.60^2 \times 100)/(0.40^2 \times 150) = 36/24 = 1.5$ ; the common resistance cancels.

[1 mark] Both charges are 60 C:  $0.40 \times 150 = 0.60 \times 100$ . But heat is not fixed by charge alone. At higher current the same coil has higher voltage, so each coulomb transfers more energy; trial B generates 1.5 times as much heat.

[1 mark] Equate  $I^2Rt$ :  $t_B = 0.40^2 \times 150/0.60^2 = 66.7$  s approximately. Use the same resistance in both trials.

### Mark allocation

$H_B/H_A = (0.60^2 \times 100)/(0.40^2 \times 150) = 36/24 = 1.5$ ; the common resistance cancels. [1 marks]

Both charges are 60 C:  $0.40 \times 150 = 0.60 \times 100$ . But heat is not fixed by charge alone. At higher current the same coil has higher voltage, so each coulomb transfers more energy; trial B generates 1.5 times as much heat. [1 marks]

Equate  $I^2Rt$ :  $t_B = 0.40^2 \times 150/0.60^2 = 66.7$  s approximately. Use the same resistance in both trials. [1 marks]

### Solution 31 — special-39-01-q31

[1 mark]  $R = V^2/P = 12^2/6 = 24 \Omega$ .

[1 mark] The claim is false under the constant-R assumption.  $P_{new} = 6^2/24 = 1.5$  W, one-quarter of 6 W, because power varies as  $V^2$  at fixed R.

[1 mark] No. A real filament's changing temperature changes its resistance, and perceived brightness is not specified by this constant-resistance electrical model. The 1.5 W result applies to the given model, not an exact brightness prediction.

### Mark allocation

$R = V^2/P = 12^2/6 = 24 \Omega$ . [1 marks]

The claim is false under the constant-R assumption.  $P_{new} = 6^2/24 = 1.5$  W, one-quarter of 6 W, because power varies as  $V^2$  at fixed R. [1 marks]

No. A real filament's changing temperature changes its resistance, and perceived brightness is not specified by this constant-resistance electrical model. The 1.5 W result applies to the given model, not an exact brightness prediction. [1 marks]

### Solution 32 — special-39-01-q32

[1 mark] Lamp:  $0.040 \times 1.5 = 0.060$  kWh. Heater:  $0.200 \times 0.30 = 0.060$  kWh. Total = 0.120 kWh.

[1 mark] No: 0.36 kWh is three times the expected 0.12 kWh, an excess of 0.24 kWh, for the actual stated operation.

[1 mark] Using 1.5 h incorrectly for the heater gives  $0.200 \times 1.5 + 0.060 = 0.360$  kWh, matching the recorded total exactly. The accounting error explains the number without requiring any electrical fault.

### Mark allocation

Lamp:  $0.040 \times 1.5 = 0.060$  kWh. Heater:  $0.200 \times 0.30 = 0.060$  kWh. Total = 0.120 kWh. [1 marks]

No: 0.36 kWh is three times the expected 0.12 kWh, an excess of 0.24 kWh, for the actual stated operation. [1 marks]

Using 1.5 h incorrectly for the heater gives  $0.200 \times 1.5 + 0.060 = 0.360$  kWh, matching the recorded total exactly. The accounting error explains the number without requiring any electrical fault. [1 marks]

### Solution 33 — special-39-01-q33

[1 mark]  $\rho = RA/L$ , so  $\rho_Y/\rho_X = (8 \times 3A/(2L))/(8 \times A/L) = 3/2$ .

[1 mark] No. The measured equality of R does not imply equal resistivity. Here Y has 1.5 times X's resistivity at the same temperature, so the evidence contradicts an equal-resistivity identification.

[1 mark] Prepare Y with length L and area A, matching X, and keep both at the same temperature. Then  $R_Y/R_X = \rho_Y/\rho_X$  directly. Also accept any new geometry with the same length-to-area ratio L/A, such as length 2L and area 2A, with the same temperature. Matching only one dimension while leaving the other unequal is insufficient.

### Mark allocation

$\rho = RA/L$ , so  $\rho_Y/\rho_X = (8 \times 3A/(2L))/(8 \times A/L) = 3/2$ . [1 marks]

No. The measured equality of R does not imply equal resistivity. Here Y has 1.5 times X's resistivity at the same temperature, so the evidence contradicts an equal-resistivity identification. [1 marks]

Prepare Y with length L and area A, matching X, and keep both at the same temperature. Then  $R_Y/R_X = \rho_Y/\rho_X$  directly. Also accept any new geometry with the same length-to-area ratio L/A, such as length 2L and area 2A, with the same temperature. Matching only one dimension while leaving the other unequal is insufficient. [1 marks]

### Solution 34 — special-39-01-q34

[1 mark] At 0 V, where actual current is zero, the graph reads 0.10 A. The zero offset is +0.10 A; subtract 0.10 A from every indication.

[1 mark] At 2 V and 6 V the indications are 0.30 A and 0.70 A.  $\Delta V = 4$  V and  $\Delta I = 0.40$  A, giving

$$R = \Delta V / \Delta I = 4 / 0.40 = 10 \, \Omega.$$

[1 mark] Each indication contains the same extra 0.10 A. In taking the difference that extra amount is subtracted from itself and disappears. Therefore the change in indicated current equals the change in true current. Equivalently, corrected currents 0.20 A and 0.60 A both give 10  $\Omega$ .

[1 mark] The uncorrected result is  $2 / 0.30 = 6.67 \, \Omega$  approximately, too small. The numerator is correct but the denominator includes 0.10 A that is not actual current; a ratio from the origin is inappropriate for the shifted line.

[1 mark] True current at 3 V is  $3 / 10 = 0.30$  A. The unadjusted meter adds 0.10 A, so it indicates 0.40 A.

### Mark allocation

At 0 V, where actual current is zero, the graph reads 0.10 A. The zero offset is +0.10 A; subtract 0.10 A from every indication. [1 marks]

At 2 V and 6 V the indications are 0.30 A and 0.70 A.  $\Delta V = 4$  V and  $\Delta I = 0.40$  A, giving  $R = \Delta V / \Delta I = 4 / 0.40 = 10 \, \Omega$ . [1 marks]

Each indication contains the same extra 0.10 A. In taking the difference that extra amount is subtracted from itself and disappears. Therefore the change in indicated current equals the change in true current. Equivalently, corrected currents 0.20 A and 0.60 A both give 10  $\Omega$ . [1 marks]

The uncorrected result is  $2 / 0.30 = 6.67 \, \Omega$  approximately, too small. The numerator is correct but the denominator includes 0.10 A that is not actual current; a ratio from the origin is inappropriate for the shifted line. [1 marks]

True current at 3 V is  $3 / 10 = 0.30$  A. The unadjusted meter adds 0.10 A, so it indicates 0.40 A. [1 marks]

### Solution 35 — special-39-01-q35

[1 mark] Q and R isolate length: they have the same alloy, temperature and diameter 2d, while Q has length 2L and R has length L.

[1 mark] P and R isolate diameter: both have length L, the same material and temperature. Diameters are d and 2d, so  $A_R / A_P = (2d/d)^2 = 4$ , not 2.

[1 mark] For P,  $R_P = 1.2 / 0.12 = 10 \, \Omega$ . Q has twice the length and four times the area of P, so  $R_Q / R_P = 2/4 = 1/2$ .  $R_Q = 5 \, \Omega$  and  $I_Q = 1.2 / 5 = 0.24$  A.

[1 mark] R has the same length and four times the area of P, so  $R_R = R_P / 4 = 2.5 \, \Omega$ .  $I_R = 1.2 / 2.5 = 0.48$  A.

[1 mark] P and Q differ in diameter as well as length. The area increase outweighs the length increase. With area held constant, doubling length doubles resistance; the halved resistance in this pair is a combined-geometry effect, not the isolated length effect.

### Mark allocation

Q and R isolate length: they have the same alloy, temperature and diameter 2d, while Q has length 2L and R has length L. [1 marks]

P and R isolate diameter: both have length L, the same material and temperature. Diameters are d and 2d, so  $A_R / A_P = (2d/d)^2 = 4$ , not 2. [1 marks]

For P,  $R_P = 1.2/0.12 = 10\ \Omega$ . Q has twice the length and four times the area of P, so  $R_Q/R_P = 2/4 = 1/2$ .  $R_Q = 5\ \Omega$  and  $I_Q = 1.2/5 = 0.24\ \text{A}$ . [1 marks]

R has the same length and four times the area of P, so  $R_R = R_P/4 = 2.5\ \Omega$ .  $I_R = 1.2/2.5 = 0.48\ \text{A}$ . [1 marks]

P and Q differ in diameter as well as length. The area increase outweighs the length increase. With area held constant, doubling length doubles resistance; the halved resistance in this pair is a combined-geometry effect, not the isolated length effect. [1 marks]

### Solution 36 — special-39-01-q36

[1 mark] The standard's measured resistance is  $V_{\text{resistor}}/I = 4.0/0.40 = 10\ \Omega$ . It has not changed under the stated alternatives.

[1 mark] Choose B. The contact must account for the source voltage not appearing across the standard:  $V_{\text{contact}} = 6 - 4 = 2\ \text{V}$ .

[1 mark] The same  $0.40\ \text{A}$  flows through the contact, so  $R_{\text{contact}} = 2/0.40 = 5\ \Omega$ . Check: total resistance  $15\ \Omega$  gives  $6/15 = 0.40\ \text{A}$ .

[1 mark] Place the ideal voltmeter across the two sides of the suspect contact, keeping the series ammeter in place. The faulty contact reads  $2\ \text{V}$ ; a sound zero-resistance contact reads  $0\ \text{V}$ . Probe contacts in turn if its position is not yet known.

[1 mark]  $P_{\text{contact}} = I^2 R_{\text{contact}} = 0.40^2 \times 5 = 0.80\ \text{W}$ , also  $2\ \text{V} \times 0.40\ \text{A}$ . This is electrical heat generation, not a temperature-rise prediction.

### Mark allocation

The standard's measured resistance is  $V_{\text{resistor}}/I = 4.0/0.40 = 10\ \Omega$ . It has not changed under the stated alternatives. [1 marks]

Choose B. The contact must account for the source voltage not appearing across the standard:  $V_{\text{contact}} = 6 - 4 = 2\ \text{V}$ . [1 marks]

The same  $0.40\ \text{A}$  flows through the contact, so  $R_{\text{contact}} = 2/0.40 = 5\ \Omega$ . Check: total resistance  $15\ \Omega$  gives  $6/15 = 0.40\ \text{A}$ . [1 marks]

Place the ideal voltmeter across the two sides of the suspect contact, keeping the series ammeter in place. The faulty contact reads  $2\ \text{V}$ ; a sound zero-resistance contact reads  $0\ \text{V}$ . Probe contacts in turn if its position is not yet known. [1 marks]

$P_{\text{contact}} = I^2 R_{\text{contact}} = 0.40^2 \times 5 = 0.80\ \text{W}$ , also  $2\ \text{V} \times 0.40\ \text{A}$ . This is electrical heat generation, not a temperature-rise prediction. [1 marks]

### Solution 37 — special-39-01-q37

[1 mark] No current flows from  $20\ \text{s}$  to  $40\ \text{s}$  because cumulative charge stays at  $8\ \text{C}$ ; no additional charge crosses the section.

[1 mark] First rising interval:  $I = (8 - 0)/(20 - 0) = 0.40\ \text{A}$ . Final rising interval:  $I = (16 - 8)/(50 - 40) = 0.80\ \text{A}$ . Award 0.5 for each.

[1 mark] The claim is false:  $P_{\text{first}} = 3 \times 0.40 = 1.2\ \text{W}$  and  $P_{\text{last}} = 3 \times 0.80 = 2.4\ \text{W}$ . Equal charge was transferred in different durations, so powers differ.

[1 mark] The total energy in each rising interval is nevertheless equal:  $E = V\Delta Q = 3 \times 8 = 24\ \text{J}$ . Checks:  $1.2 \times 20 = 2.4 \times 10 = 24\ \text{J}$ . Equal energy is not equal power.

**Mark allocation**

No current flows from 20 s to 40 s because cumulative charge stays at 8 C; no additional charge crosses the section. [1 marks]

First rising interval:  $I = (8 - 0)/(20 - 0) = 0.40$  A. Final rising interval:  $I = (16 - 8)/(50 - 40) = 0.80$  A. Award 0.5 for each. [1 marks]

The claim is false:  $P_{\text{first}} = 3 \times 0.40 = 1.2$  W and  $P_{\text{last}} = 3 \times 0.80 = 2.4$  W. Equal charge was transferred in different durations, so powers differ. [1 marks]

The total energy in each rising interval is nevertheless equal:  $E = V\Delta Q = 3 \times 8 = 24$  J. Checks:  $1.2 \times 20 = 2.4 \times 10 = 24$  J. Equal energy is not equal power. [1 marks]

**Solution 38 — special-39-01-q38**

[1 mark]  $R_X = 3/0.30 = 10\ \Omega$  and  $R_Y = 3/0.15 = 20\ \Omega$ . Award 0.5 for each.

[1 mark]  $R_{\text{box}} = 6/0.20 = 30\ \Omega$ , which equals  $10 + 20$ . The box is series, under the two allowed alternatives.

[1 mark] For parallel at 6 V,  $I_X = 6/10 = 0.60$  A and  $I_Y = 6/20 = 0.30$  A; total = 0.90 A, not the measured 0.20 A.

[1 mark] Parallel branches have equal voltage, not necessarily equal current. Unequal resistances give unequal branch currents (here in ratio 2:1), so equal splitting cannot rescue the proposed parallel explanation.

**Mark allocation**

$R_X = 3/0.30 = 10\ \Omega$  and  $R_Y = 3/0.15 = 20\ \Omega$ . Award 0.5 for each. [1 marks]

$R_{\text{box}} = 6/0.20 = 30\ \Omega$ , which equals  $10 + 20$ . The box is series, under the two allowed alternatives. [1 marks]

For parallel at 6 V,  $I_X = 6/10 = 0.60$  A and  $I_Y = 6/20 = 0.30$  A; total = 0.90 A, not the measured 0.20 A. [1 marks]

Parallel branches have equal voltage, not necessarily equal current. Unequal resistances give unequal branch currents (here in ratio 2:1), so equal splitting cannot rescue the proposed parallel explanation. [1 marks]

**Solution 39 — special-39-01-q39**

[1 mark] Electrical energy  $E = VIt = 5 \times 0.40 \times 120 = 240$  J. Energy not captured =  $240 - 180 = 60$  J. Award 0.5 for each.

[1 mark] No. The electrical readings give  $R = V/I = 12.5\ \Omega$ . The collector measures only the captured portion, not all generated heat. The 60 J transferred to the surroundings is an allowed explanation; lower collection does not diagnose wrong resistance.

[1 mark] Electrical heat generated stays 240 J because  $V$ ,  $I$  and  $t$  are unchanged. Better insulation does not change that electrical input under the stated conditions.

[1 mark] The captured energy increases by  $225 - 180 = 45$  J. Heat going to the surroundings falls from 60 J to  $240 - 225 = 15$  J, a decrease of 45 J. This is redistribution of the same generated heat, not additional generation.

**Mark allocation**

Electrical energy  $E = VIt = 5 \times 0.40 \times 120 = 240 \text{ J}$ . Energy not captured  $= 240 - 180 = 60 \text{ J}$ . Award 0.5 for each. [1 marks]

No. The electrical readings give  $R = V/I = 12.5 \Omega$ . The collector measures only the captured portion, not all generated heat. The 60 J transferred to the surroundings is an allowed explanation; lower collection does not diagnose wrong resistance. [1 marks]

Electrical heat generated stays 240 J because  $V$ ,  $I$  and  $t$  are unchanged. Better insulation does not change that electrical input under the stated conditions. [1 marks]

The captured energy increases by  $225 - 180 = 45 \text{ J}$ . Heat going to the surroundings falls from 60 J to  $240 - 225 = 15 \text{ J}$ , a decrease of 45 J. This is redistribution of the same generated heat, not additional generation. [1 marks]

## Source-pattern and rationale appendix

These questions are original. References identify supplied source patterns and curriculum connections, not official provenance for these questions or predictions of future examinations. No missing references or rationale have been inferred.

### Question 1 — special-39-01-q01

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172

Competency classification: No

Competency marks: 0

Competency rationale: Direct unit conversion and substitution, not an evidence-based decision.

Rationale tags: foundational

### Question 2 — special-39-01-q02

Source patterns: ncert-11

NCERT references: §11.2, printed p.173

Competency classification: No

Competency marks: 0

Competency rationale: Direct use of the definition  $V = W/Q$ .

Rationale tags: foundational

### Question 3 — special-39-01-q03

Source patterns: ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–180

Competency classification: No

Competency marks: 0

Competency rationale: Foundational dimensional interpretation.

Rationale tags: foundational

### Question 4 — special-39-01-q04

Source patterns: cbse\_sqp\_2017-18\_electricity\_q26  
pyq-2023-246eac3017f0793c27bba6a6  
ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176  
§11.5 / Activity 11.3, printed pp.177–180

Competency classification: Yes

Competency marks: 1

Competency rationale: Requires reading the supplied inverted axes and linking resistance to resistivity under controlled geometry.

Rationale tags: evidence-reasoning

### Question 5 — special-39-01-q05

Source patterns: sqp\_2022-23\_q13  
ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176  
§11.5 / Activity 11.3, printed pp.177–180

Competency classification: Yes

Competency marks: 1

Competency rationale: Selects a controlled follow-up to a confounded observation rather than discarding data.

Rationale tags: evidence-reasoning

### Question 6 — special-39-01-q06

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172  
§11.7, printed pp.188–189

Competency classification: No

Competency marks: 0

Competency rationale: Conservative audit: the equal-current observation merely cues the familiar charge-versus-energy distinction; count as foundational recall.

Rationale tags: foundational

### Question 7 — special-39-01-q07

Source patterns: ncert-11

NCERT references: §11.2, printed p.173  
§11.6.1 / Activities 11.4–11.5, printed pp.182–184

Competency classification: Yes

Competency marks: 1

Competency rationale: Must trace the pictured meter endpoints and distinguish the measured interval from the intended one.

Rationale tags: evidence-reasoning

### Question 8 — special-39-01-q08

Source patterns: pyq-2018-7253f82c5332b612f2b92314  
cbse\_sqp\_2017-18\_electricity\_q26  
ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 1

Competency rationale: Applies an independently observed zero error and checks consistency across two settings.

Rationale tags: evidence-reasoning

### Question 9 — special-39-01-q09

Source patterns: cbse\_sqp\_2018-19\_electricity\_q13  
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–180

Competency classification: Yes

Competency marks: 1

Competency rationale: Chooses controls necessary for a causal material comparison.

Rationale tags: evidence-reasoning

### Question 10 — special-39-01-q10

Source patterns: pyq-2024-1e2d386b35acad1e9e07f369  
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–180

Competency classification: No

Competency marks: 0

Competency rationale: Standard proportionality recall; no experimental decision is needed.

Rationale tags: foundational

### Question 11 — special-39-01-q11

Source patterns: sqp\_2020-21\_q9\_main  
ncert-11

NCERT references: §11.6.2 / Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 1

Competency rationale: Uses a limiting bound to reject evidence, rather than only substituting into a formula.

Rationale tags: evidence-reasoning

### Question 12 — special-39-01-q12

Source patterns: SQP\_2023-24\_q32  
ncert-11

NCERT references: §11.7, printed pp.188–189

Competency classification: No

Competency marks: 0

Competency rationale: Conservative audit: direct substitution of two supplied scale factors into  $H = I^2Rt$ ; no independent evidence diagnosis or experimental decision.

Rationale tags: foundational

### Question 13 — special-39-01-q13

Source patterns: ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: No

Competency marks: 0

Competency rationale: Direct conversion of a power-time product.

Rationale tags: foundational

### **Question 14 — special-39-01-q14**

Source patterns: SQP\_2023-24\_q32  
ncert-11

NCERT references: §11.7, printed pp.188–189  
§11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 1

Competency rationale: Identifies the hidden fixed-current assumption and selects the correct controlled-variable relation.

Rationale tags: evidence-reasoning

### **Question 15 — special-39-01-q15**

Source patterns: sqp\_2020-21\_q9\_main  
ncert-11

NCERT references: §11.6.2 / Activity 11.6, printed pp.185–186  
§11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Distinguishes branch behaviour from total behaviour after a circuit change.

Rationale tags: evidence-reasoning

### **Question 16 — special-39-01-q16**

Source patterns: ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: No

Competency marks: 0

Competency rationale: Foundational definition, deliberately not counted as competency.

Rationale tags: foundational

### **Question 17 — special-39-01-q17**

Source patterns: pyq-2018-7253f82c5332b612f2b92314  
cbse\_sqp\_2017-18\_electricity\_q26  
ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 1

Competency rationale: Evaluates an alternative measurement explanation for apparent non-ohmic evidence.

Rationale tags: evidence-reasoning

### **Question 18 — special-39-01-q18**

Source patterns: cbse\_sqp\_2018-19\_electricity\_q13  
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–180

Competency classification: Yes

Competency marks: 1

Competency rationale: Tests whether a material claim follows from an incomplete measurement.

Rationale tags: evidence-reasoning

### **Question 19 — special-39-01-q19**

Source patterns: sqp\_2020-21\_q9\_main  
ncert-11

NCERT references: §11.6.1 / Activities 11.4–11.5, printed pp.182–184  
§11.6.2 / Activity 11.6, printed pp.185–186

Competency classification: No

Competency marks: 0

Competency rationale: Tests familiar circuit properties and their explanatory relation without a new evidence task.

Rationale tags: foundational

### **Question 20 — special-39-01-q20**

Source patterns: ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: No

Competency marks: 0

Competency rationale: Basic unit knowledge, not contextual reasoning.

Rationale tags: foundational

### **Question 21 — special-39-01-q21**

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172

Competency classification: Yes

Competency marks: 1

Competency rationale: Only the evidence-based rejection in (ii) is counted; two direct calculations in (i) are foundational.

Rationale tags: evidence-reasoning  
partial-credit-classification

### Question 22 — special-39-01-q22

Source patterns: ncert-11

NCERT references: §11.2, printed p.173

Competency classification: Yes

Competency marks: 1

Competency rationale: The cross-record voltage inference in (i) is competency; the subsequent direct prediction is classified conservatively as foundational.

Rationale tags: evidence-reasoning  
partial-credit-classification

### Question 23 — special-39-01-q23

Source patterns: pyq-2018-7253f82c5332b612f2b92314  
ncert-11

NCERT references: §11.1, printed p.172: ammeter measures current and is connected in series  
Activity 11.2, printed pp.176–177: ammeter with specified 0–5 A range and component-current observations

Competency classification: Yes

Competency marks: 2

Competency rationale: Two constrained laboratory decisions, one mark each: (i) select a meter that simultaneously accommodates the supplied current interval and resolves the required change; (ii) reject a range-only workaround when neither meter meets both constraints and specify a suitable replacement. Arithmetic supports these decisions and earns no stand-alone mark. No wire-control rehearsal or zero-offset correction is required.

Rationale tags: evidence-reasoning  
instrument-selection  
range-resolution-tradeoff  
constraint-evaluation

### Question 24 — special-39-01-q24

Source patterns: cbse\_sqp\_2018-19\_electricity\_q13  
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–180

Competency classification: Yes

Competency marks: 1

Competency rationale: Only (ii), distinguishing material property from specimen resistance, earns competency credit.

Rationale tags: evidence-reasoning  
partial-credit-classification

### Question 25 — special-39-01-q25

Source patterns: pyq-2018-7253f82c5332b612f2b92314  
ncert-11

NCERT references: §11.6.1 / Activities 11.4–11.5, printed pp.182–184  
§11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 2

Competency rationale: Uses the series sum and independent current evidence to identify a common measurement error.

Rationale tags: evidence-reasoning

### **Question 26 — special-39-01-q26**

Source patterns: sqp\_2020-21\_q9\_main  
ncert-11

NCERT references: §11.6.2 / Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 1

Competency rationale: Only (ii), testing the proposed resistance label against branch evidence, is counted. The direct current subtraction in (i) is foundational.

Rationale tags: evidence-reasoning  
partial-credit-classification

### **Question 27 — special-39-01-q27**

Source patterns: cbse\_sqp\_2017-18\_electricity\_q26  
sqp\_2022-23\_q13  
ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176  
§11.5 / Activity 11.3, printed pp.177–180

Competency classification: Yes

Competency marks: 2

Competency rationale: Only (ii) and (iii), evaluating constant resistance and designing a controlled repeat, are counted. Graph reading and V/I calculations in (i) are excluded conservatively.

Rationale tags: evidence-reasoning  
partial-credit-classification

### **Question 28 — special-39-01-q28**

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172  
§11.2, printed p.173  
§11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 3

Competency rationale: Requires tracing both faults and their interaction, not simply naming instrument conventions.

Rationale tags: evidence-reasoning

### **Question 29 — special-39-01-q29**

Source patterns: sqp\_2020-21\_q9\_main  
ncert-11

NCERT references: §11.6.2 / Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 2

Competency rationale: (ii) and (iii) reason about a circuit change; routine equivalent-resistance calculation (i) is excluded.

Rationale tags: evidence-reasoning  
partial-credit-classification

### **Question 30 — special-39-01-q30**

Source patterns: SQP\_2023-24\_q32  
ncert-11

NCERT references: §11.1, printed pp.171–172  
§11.7, printed pp.188–189

Competency classification: Yes

Competency marks: 1

Competency rationale: Only (ii), rejecting equal-charge implies equal-heat reasoning using energy per coulomb, is counted. The ratio in (i) and rearrangement in (iii) are routine calculations.

Rationale tags: evidence-reasoning  
partial-credit-classification

### **Question 31 — special-39-01-q31**

Source patterns: sqp\_2022-23\_q13  
ncert-11

NCERT references: §11.8, printed pp.191–192  
§11.5 / Activity 11.3, printed pp.177–180

Competency classification: Yes

Competency marks: 2

Competency rationale: Tests an extrapolation and identifies model limits in (ii)–(iii); rated-resistance substitution is excluded.

Rationale tags: evidence-reasoning  
partial-credit-classification

### **Question 32 — special-39-01-q32**

Source patterns: ncert-11

NCERT references: §11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 1

Competency rationale: Only (iii), reconstructing the proposed timing error and checking its exact agreement, is counted. The sum and direct comparison in (i)–(ii) are foundational.

Rationale tags: evidence-reasoning  
partial-credit-classification

### Question 33 — special-39-01-q33

Source patterns: cbse\_sqp\_2018-19\_electricity\_q13  
pyq-2024-1e2d386b35acad1e9e07f369  
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–180

Competency classification: Yes

Competency marks: 2

Competency rationale: Only (ii) and (iii), evaluating the material identification and designing a direct comparison, are counted. The ratio calculation in (i) is excluded.

Rationale tags: evidence-reasoning  
partial-credit-classification

### Question 34 — special-39-01-q34

Source patterns: cbse\_sqp\_2017-18\_electricity\_q26  
pyq-2018-7253f82c5332b612f2b92314  
ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176  
§11.1, printed pp.171–172

Competency classification: Yes

Competency marks: 4

Competency rationale: Calibration from the plotted zero, difference-based removal of error and critique of the uncorrected ratio earn the first four marks. The final calibrated prediction is excluded conservatively as a direct substitution.

Rationale tags: evidence-reasoning  
partial-credit-classification

### Question 35 — special-39-01-q35

Source patterns: cbse\_sqp\_2018-19\_electricity\_q13  
pyq-2024-1e2d386b35acad1e9e07f369  
ncert-11

NCERT references: §11.5 / Activity 11.3, printed pp.177–180  
§11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 3

Competency rationale: Controlled-pair selection in (i)–(ii) and the confounding critique in (iv) earn three marks. The two numerical current predictions in (iii) are excluded conservatively.

Rationale tags: evidence-reasoning  
partial-credit-classification

### Question 36 — special-39-01-q36

Source patterns: ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176  
§11.6.1 / Activities 11.4–11.5, printed pp.182–184  
§11.7, printed pp.188–189  
§11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 3

Competency rationale: Count the unchanged-standard diagnosis in (i), selection of fault B using the missing voltage in (ii), and confirmatory probe design in (iii). Exclude the one-mark contact-resistance substitution within (ii) and the final power substitution.

Rationale tags: evidence-reasoning  
partial-credit-classification

### **Question 37 — special-39-01-q37**

Source patterns: ncert-11

NCERT references: §11.1, printed pp.171–172  
§11.2, printed p.173  
§11.8, printed pp.191–192

Competency classification: Yes

Competency marks: 2

Competency rationale: Only the two marks in (iii), distinguishing equal energy from unequal power using the two intervals, are counted. Zero-current interval reading and direct current calculations in (i)–(ii) are foundational.

Rationale tags: evidence-reasoning  
partial-credit-classification

### **Question 38 — special-39-01-q38**

Source patterns: sqp\_2020-21\_q9\_main  
pyq-2025-3c907fffc20ec05c468bbbea  
ncert-11

NCERT references: §11.4 / Activity 11.1, printed pp.175–176  
§11.6.1 / Activities 11.4–11.5, printed pp.182–184  
§11.6.2 / Activity 11.6, printed pp.185–186

Competency classification: Yes

Competency marks: 3

Competency rationale: The isolated resistance calculations are foundational. Connection inference and testing the rival branch model earn the other three marks.

Rationale tags: evidence-reasoning  
partial-credit-classification

### **Question 39 — special-39-01-q39**

Source patterns: SQP\_2023-24\_q32  
ncert-11

NCERT references: §11.7, printed pp.188–189  
§11.8, printed pp.191–192  
§11.4 / Activity 11.1, printed pp.175–176

Competency classification: Yes

Competency marks: 2

Competency rationale: Count (ii), rejecting resistance diagnosis from incomplete heat collection, and the first mark of (iii), unchanged generated heat under a controlled repeat. Exclude initial energy arithmetic and the

final one-mark collected/lost-energy subtraction.

Rationale tags: evidence-reasoning  
partial-credit-classification

## Source reference details

id: SQP\_2023-24\_q32

title: CBSE Class X Science (086) Sample Question Paper 2023-24 (English)

url: [https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2023\\_24/Science-SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf)

source\_role: design-pattern

local\_source\_record: research/sqp/question\_bank.json

question\_number: 32

session: 2023-24

exam\_type: SQP

original\_section: Section-C

language: en

pages: 5

page: 5

bank\_verified: Yes

bank\_answer\_verified: No

bank\_answer\_source\_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: cbse\_sqp\_2017-18\_electricity\_q26

title: Sample Question Paper 2017-18 — Science — Class - X

url: [https://cbseacademic.nic.in/web\\_material/SQP/CLASS\\_X\\_2017\\_18/Science\\_SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2017_18/Science_SQP.pdf)

source\_role: design-pattern

local\_source\_record: research/sqp/question\_bank.json

question\_number: 26

session: 2017-18

exam\_type: SQP

original\_section: SECTION - B

language: en

pages: 3

4

page: 3

bank\_verified: Yes

bank\_answer\_verified: No

bank\_answer\_source\_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: cbse\_sqp\_2018-19\_electricity\_q13

title: Class-X Science-086 SAMPLE QUESTION PAPER-19

url: [https://cbseacademic.nic.in/web\\_material/SQP/CLASS\\_X\\_2018\\_19/X\\_Science\\_SQP\\_2018-19.pdf](https://cbseacademic.nic.in/web_material/SQP/CLASS_X_2018_19/X_Science_SQP_2018-19.pdf)

source\_role: design-pattern

local\_source\_record: research/sqp/question\_bank.json

question\_number: 13

session: 2018-19

exam\_type: SQP

original\_section: SECTION C

language: en

pages: 3

page: 3

bank\_verified: Yes

bank\_answer\_verified: Yes

bank\_answer\_source\_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: ncert-11

title: NCERT Science, Class X, Chapter 11: Electricity

url: <https://ncert.nic.in/textbook/pdf/jesc111.pdf>

source\_role: conceptual-scope

local\_source\_record: research/sqp/archive/jesc111.txt

notes: Conceptual scope and activity reference only; not provenance of original questions. No question-bank verification flags are inferred.

id: pyq-2018-7253f82c5332b612f2b92314

title: CBSE — Class X Science — 2018 — 31/B — main

url: [https://www.cbse.gov.in/curric~1/qpms2018/qp\\_10/Science.zip](https://www.cbse.gov.in/curric~1/qpms2018/qp_10/Science.zip)

source\_role: design-pattern

local\_source\_record: research/pyq/question\_bank.json

question\_number: 27

year: 2018

session: main

exam\_type: PYQ

paper\_code: 31/B

language: English

source\_title\_generated: Yes

session\_inferred: Yes

original\_section: SECTION-B

pages: 15

page: 15

bank\_verified: Yes

bank\_answer\_verified: No

bank\_official\_source\_verified: Yes

bank\_text\_verified: Yes

bank\_visual\_verified: Yes

bank\_year\_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: pyq-2023-246eac3017f0793c27bba6a6

title: CBSE — Class X Science — 2023 — main — 31/2/1

url: <https://www.cbse.gov.in/cbsenew/question-paper/2023/X/SCIENCE.zip>

source\_role: design-pattern

local\_source\_record: research/pyq/question\_bank.json

question\_number: 25

year: 2023

session: main

exam\_type: PYQ

paper\_code: 31/2/1

language: English

source\_title\_generated: Yes

original\_section: SECTION B

pages: 15

page: 15

bank\_verified: Yes

bank\_answer\_verified: No

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.

id: pyq-2024-1e2d386b35acad1e9e07f369

title: CBSE — Class X Science — 2024 — main — 31/4/1

url: <https://www.cbse.gov.in/cbsenew/question-paper/2024/X/SCIENCE.zip>

source\_role: design-pattern  
local\_source\_record: research/pyq/question\_bank.json  
question\_number: 26  
year: 2024  
session: main  
exam\_type: PYQ  
paper\_code: 31/4/1  
language: English  
source\_title\_generated: Yes  
original\_section: SECTION B  
pages: 13  
page: 13  
bank\_verified: Yes  
bank\_answer\_verified: No  
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.  
id: pyq-2025-3c907fffc20ec05c468bbbea  
title: CBSE — Class X Science — 2025 — main — 31/3/1  
url: [https://www.cbse.gov.in/cbsenew/question-paper/2025/X/086\\_Science.zip](https://www.cbse.gov.in/cbsenew/question-paper/2025/X/086_Science.zip)  
source\_role: design-pattern  
local\_source\_record: research/pyq/question\_bank.json  
question\_number: 39  
year: 2025  
session: main  
exam\_type: PYQ  
paper\_code: 31/3/1  
language: English  
source\_title\_generated: Yes  
original\_section: SECTION E  
pages: 25  
27  
page: 25  
bank\_verified: Yes  
bank\_answer\_verified: No  
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.  
id: sqp\_2020-21\_q9\_main  
title: CBSE Class X Science (086) Sample Question Paper 2020-21  
url: [https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2020\\_21/Science-SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2020_21/Science-SQP.pdf)  
source\_role: design-pattern  
local\_source\_record: research/sqp/question\_bank.json  
question\_number: 9  
session: 2020-21  
exam\_type: SQP  
original\_section: Section A  
language: en  
pages: 2  
page: 2  
bank\_verified: Yes  
bank\_answer\_verified: Yes  
bank\_answer\_source\_verified: Yes  
notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.  
id: sqp\_2022-23\_q13  
title: CBSE Class X Science (086) Sample Question Paper 2022-23

url: [https://cbseacademic.nic.in/web\\_material/SQP/ClassX\\_2022\\_23/Science-SQP.pdf](https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-SQP.pdf)

source\_role: design-pattern

local\_source\_record: research/sqp/question\_bank.json

question\_number: 13

session: 2022-23

exam\_type: SQP

original\_section: Section A

language: en

pages: 4

page: 4

bank\_verified: Yes

bank\_answer\_verified: Yes

bank\_answer\_source\_verified: Yes

notes: Design-pattern reference only, not provenance of original wording or answers. Bank verification flags describe the source record only and are not inherited by original questions.